

Robert Brown

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SUMMARY

Machine Learning Engineer with 10+ years across data science, ML, and quantitative roles, currently focused on LLM evaluation, RAG and agentic systems, and statistical measurement for high-stakes decision workflows. Led evaluation-driven LLM development at Atropos Health (ChatRWD, US patent filed), including multi-stage fine-tuning with a foundation model partner and rubric-based evaluation across HealthBench, MedHELM, and CPC. Career spans LLM systems, applied statistics, public health data leadership, and production ML, with a current focus on empirical AI safety.

SKILLS

Programming: Python, SQL, PySpark, R

Statistical & Analytical Methods: Experimentation, regression, observational analysis, metric design, evaluation methodology

LLM Systems & Evaluation: Fine-tuning, RAG, agentic workflows, rubric-based evaluation, benchmark design, HITL review

ML & Data Tooling: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Hugging Face, LangChain, LangGraph, MCP

Production Data & Infrastructure: AWS (SageMaker, Lambda, EC2, S3, OpenSearch), GCP, MLflow, Airflow, Docker, CI/CD

EXPERIENCE

FOUNDER & ML ENGINEER | STUDIO BOKEH AI, LLC (STUDIOBOKEH.AI) 2025 - Present • San Francisco, CA

- Founded an applied AI company and built an AI fitness application launching in 2026, owning the full ML stack: fine-tuned local models for local inference, deep learning pipelines, and AWS backend for training/evals, and serving
- Designed end-to-end model development and evaluation workflow as a solo technical founder, from model selection and fine-tuning through production deployment and monitoring

STAFF MACHINE LEARNING ENGINEER | ATROPOS HEALTH 2022 - May 2026 • San Francisco, CA

- Led development of ChatRWD, an evaluation-driven LLM application using retrieval-augmented generation and agentic workflows for high-stakes research and decision-support use cases (US Patent Application 20250078969 A1)
- Led a multi-stage LLM evaluation and fine-tuning initiative with a foundation model partner, establishing benchmarked baselines across HealthBench, MedHELM, and CPC and measuring statistically meaningful quality improvements across iterations
- Designed rubric-based evaluation frameworks and built scalable production pipelines for model evaluation, retrieval, and monitoring, combining expert-curated datasets and confidence-interval analysis to improve measurement quality and enable rapid iteration
- Partnered across product, engineering, research, and clinical teams to translate ambiguous real-world workflows into measurable objectives, evaluation criteria, and shipped system improvements

EPIDEMIOLOGIST II/DATA CHIEF | ALAMEDA CO. PUBLIC HEALTH DEPT. 2018 - 2021 • Oakland, CA

- Served as Data Chief for the COVID-19 Incident Command response, directing 25+ data professionals and owning end-to-end surveillance and reporting systems for 70+ communicable diseases under high-stakes, time-pressured decision-making

DATA ENGINEER | BERLIN BRANDS GROUP 2021 - 2022 • San Francisco, CA

- Built and productionized forecasting, analytics, and decision-support systems on AWS, translating business questions into measurable metrics and executive-facing outputs

DATA SCIENTIST | LOS ANGELES CO. PUBLIC HEALTH DEPT. 2021 • Los Angeles, CA

- Led production data and modeling systems supporting large-scale public-sector decision-making under uncertainty

HEALTH DATA SCIENCE FELLOW | INSIGHT FELLOWS 2020 • San Francisco, CA

- Built and evaluated predictive ML models (78% ROC AUC) for healthcare customer retention, translating results into recommendations for executive stakeholders

EDUCATION

UC BERKELEY | MPH IN BIOSTATISTICS & EPIDEMIOLOGY 2017 • Berkeley, CA

UC DAVIS | BS IN NEUROBIOLOGY, PHYSIOLOGY, & BEHAVIOR, BA IN PSYCHOLOGY 2011 • Davis, CA

RESEARCH & IP

PATENT

- "Systems and Methods for Automated Evidence Generation" - US Patent Application 20250078969 A1 (Filed 8/29/2023)

SELECTED PUBLICATIONS

- Low, Y. S., Jackson, M. L., Hyde, R. J., Brown, R. E., et al. (2025). Answering real-world clinical questions using large language model, retrieval-augmented generation, and agentic systems. DIGITAL HEALTH, 11.
- Brown, R. E., Turner, C., Hern, J., & Santos, G.M. (2017). Partner-level substance use associated with increased sexual risk behaviors among men who have sex with men in San Francisco, CA. Drug and Alcohol Dependence, 176–180.